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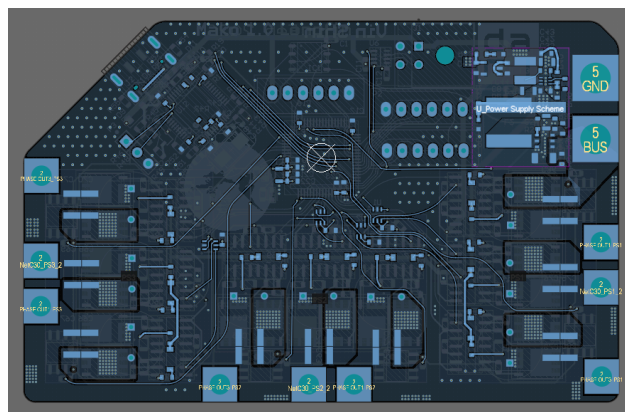
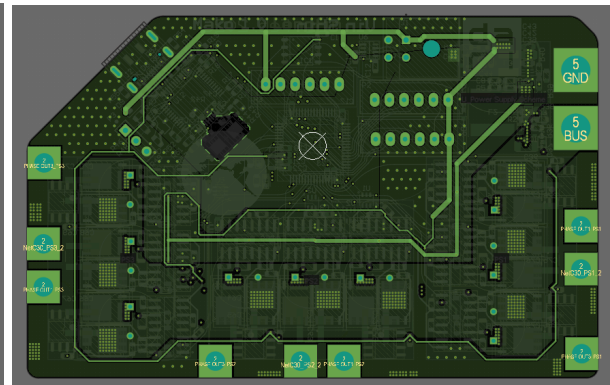
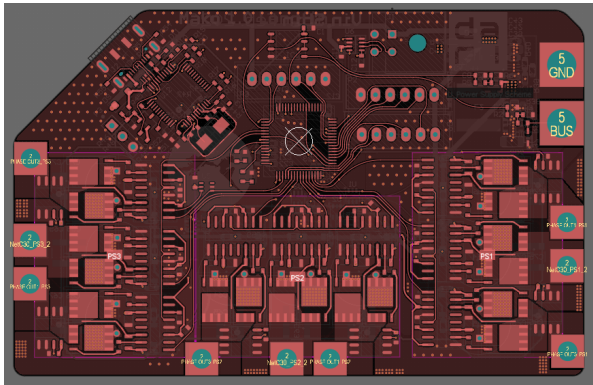
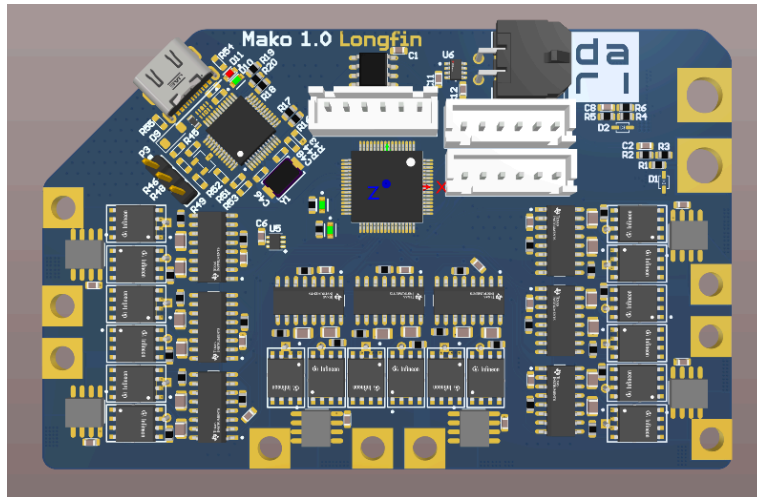
About Me

Hi y'all! I am a second year electrical engineering student at UCSB. I am a LV Electrical Lead at Gaucho Racing, UCSB's collegiate chapter of SAE. I really like making random things, which has been power electronics and RF electronics recently. But I have some experience with designing with mixed signal designs.

Here's a few stuff that I have done recently:

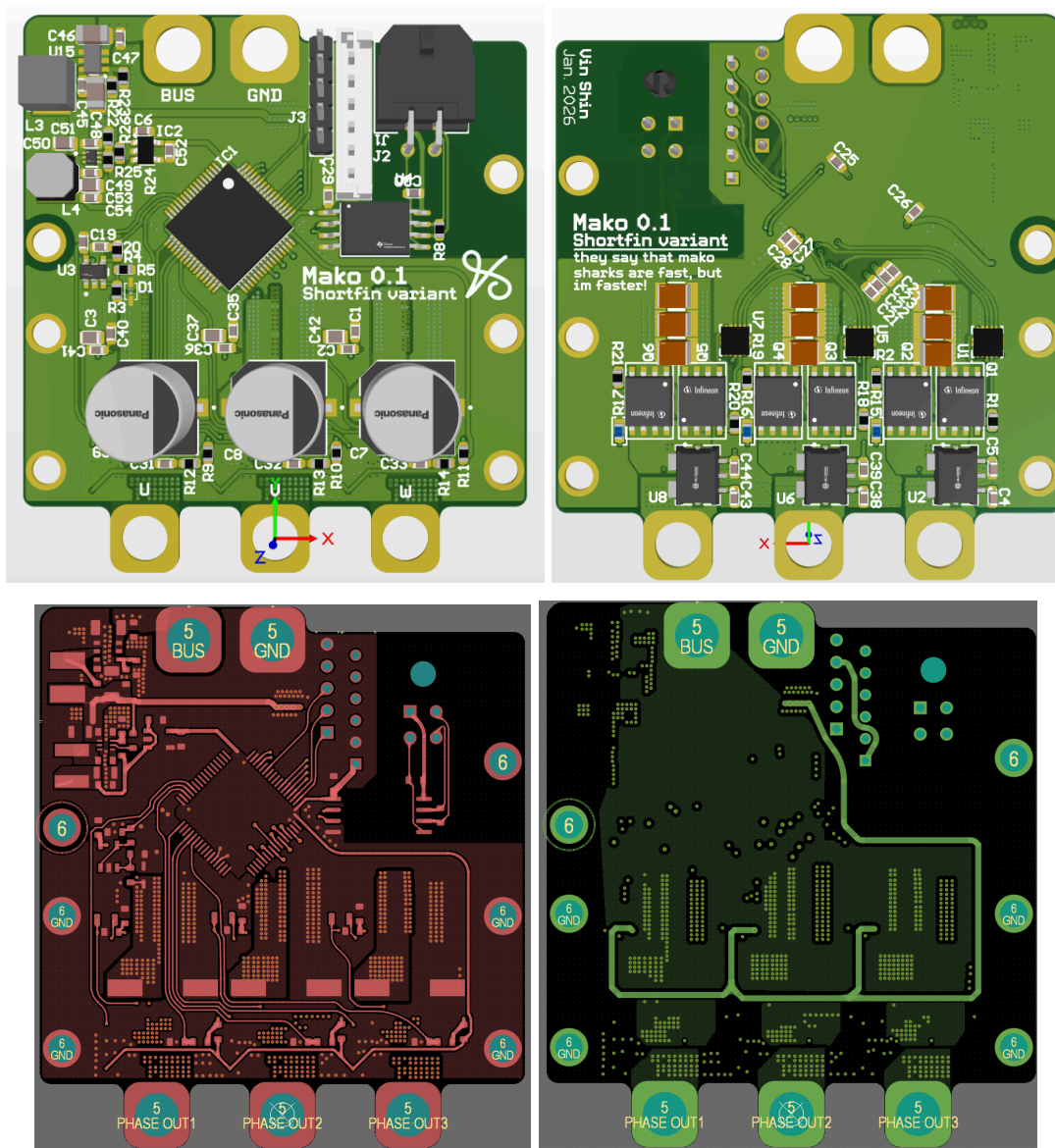
Mako Longfin - 4.5kW 3 motor inverter

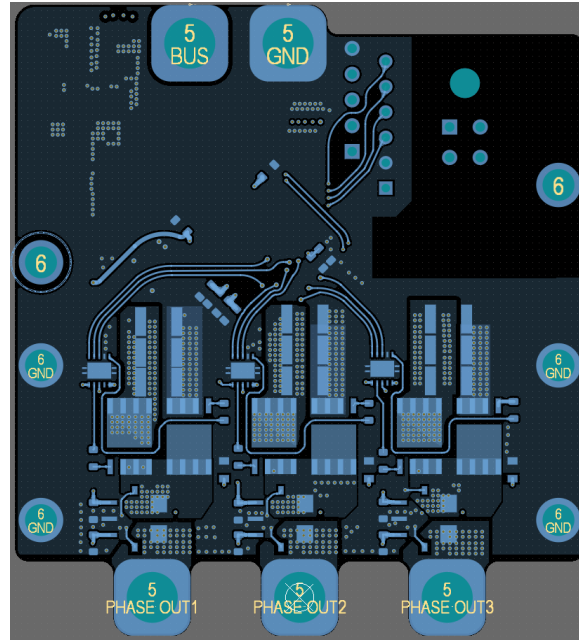
As a direct next revision of my previous motor controller, Mako Shortfin, this motor controller is designed with some tweaks that enable FOC control over 3 motors with one MCU.



Mako Shortfin - A 1.5kW FOC Motor Controller

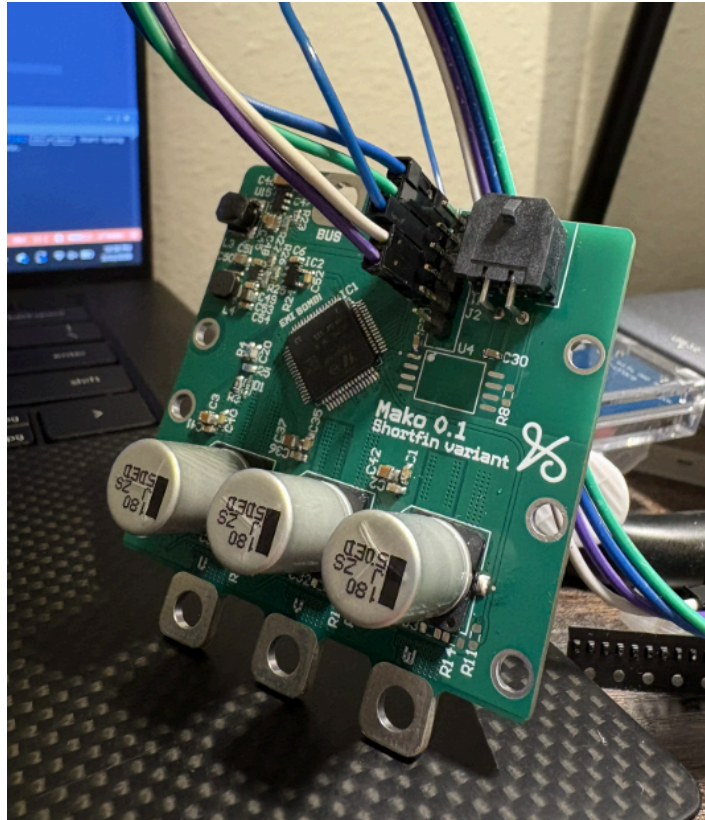
I designed a roughly 4900mm² 4 layer board with a discrete powerstage, STM32G4, power conversion, and CAN Transceiver that is operated at 30KHz switching frequency, up to a 60V DC Bus, and at 25A continuous.





(Red is Top, green is 3rd, blue is bottom. the 2nd layer is GND only, so no image is provided)

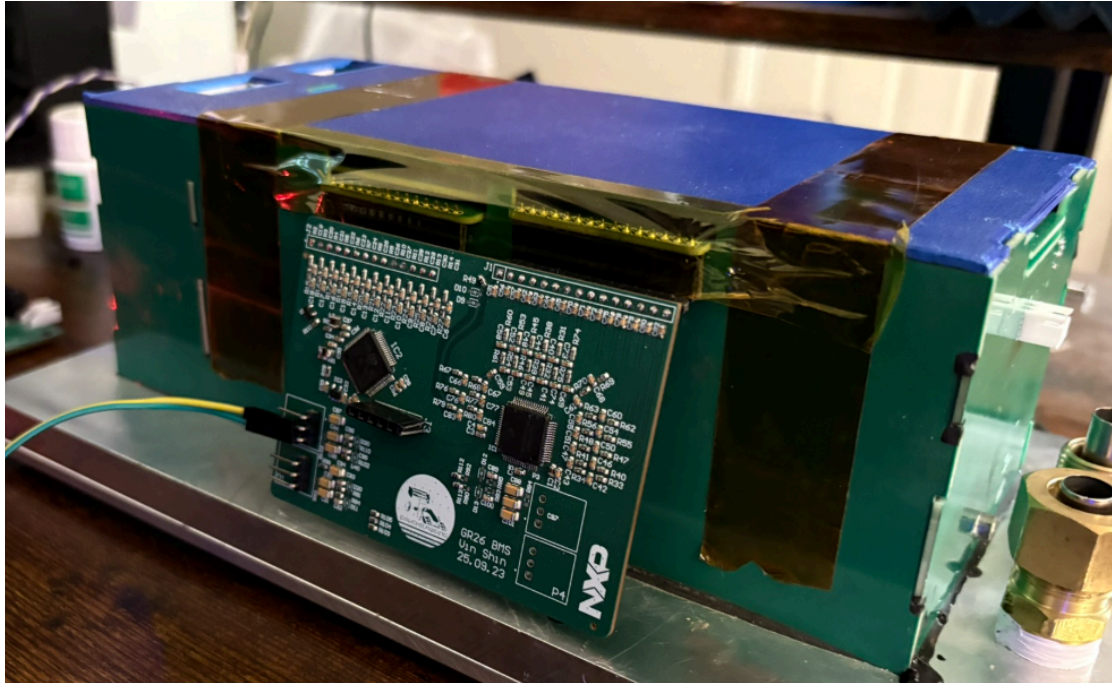
I used the newest Infineon OptiMOS MOSFETs because of their low $R_{DS,ON}$ value despite its silicon semiconductor topology, pairing with TI's UCC27302A gate drivers. The gate drivers operate in 12V, so my power conversion topology is Bus voltage to 12V to 5V to 3.3V, Buck to Buck to LDO. There are no 5V applications on my board, but my encoder sensors will require 5V so I am enabling it for external use.



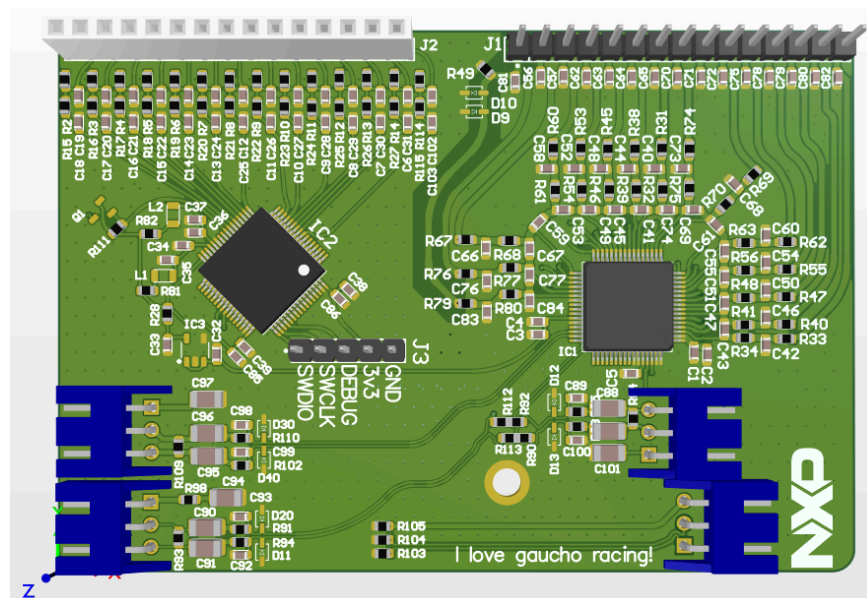
Here is the board fully assembled. Here is a [video](#) of it running in open-loop mode (no internal feedback)

GR26 Battery Management System

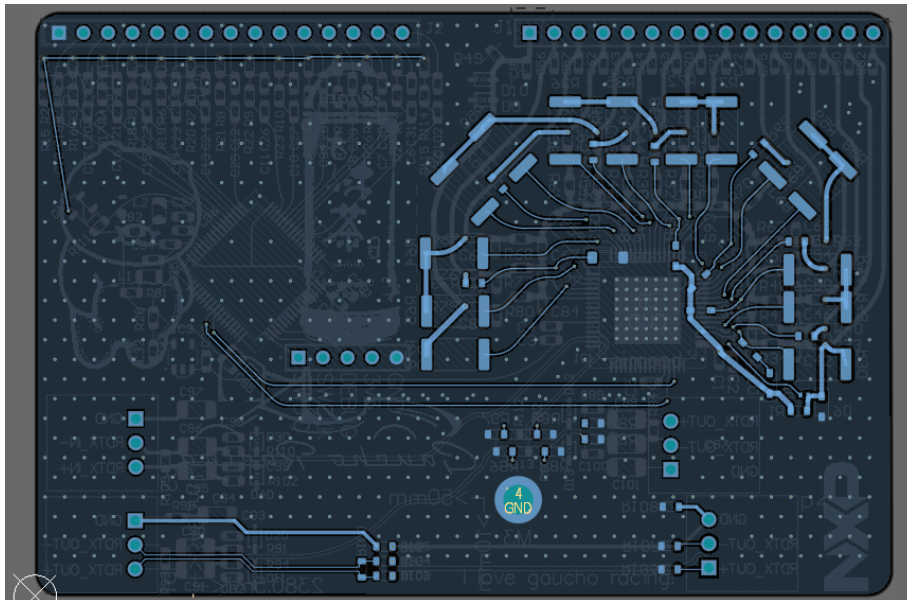
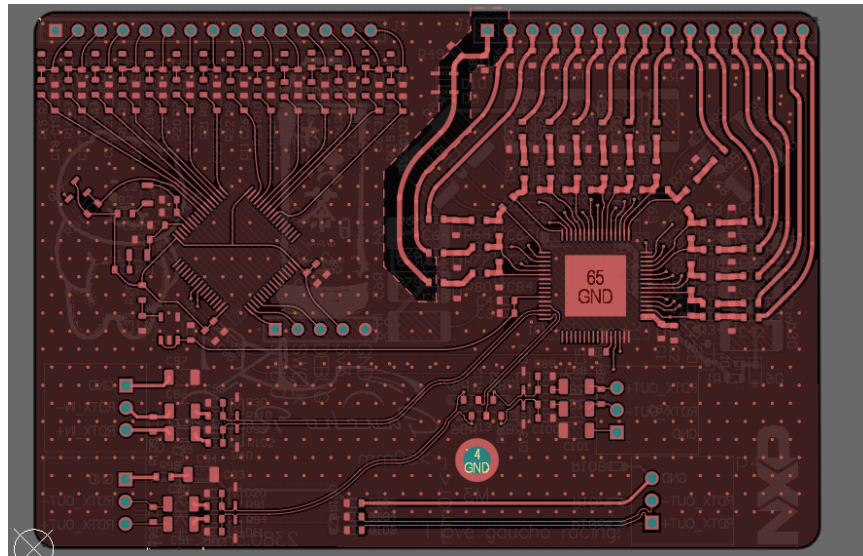
Entering Gaucho Racing's next season of 2025-2026, we decided to push for another design cycle on our main high voltage battery pack. Holding an interest in accumulator development, I had taken on the next revision of the battery pack's BMS module in our 144S2P configuration battery pack (588V).



Revision 1 shown in image above.



Revision 2 shown above.

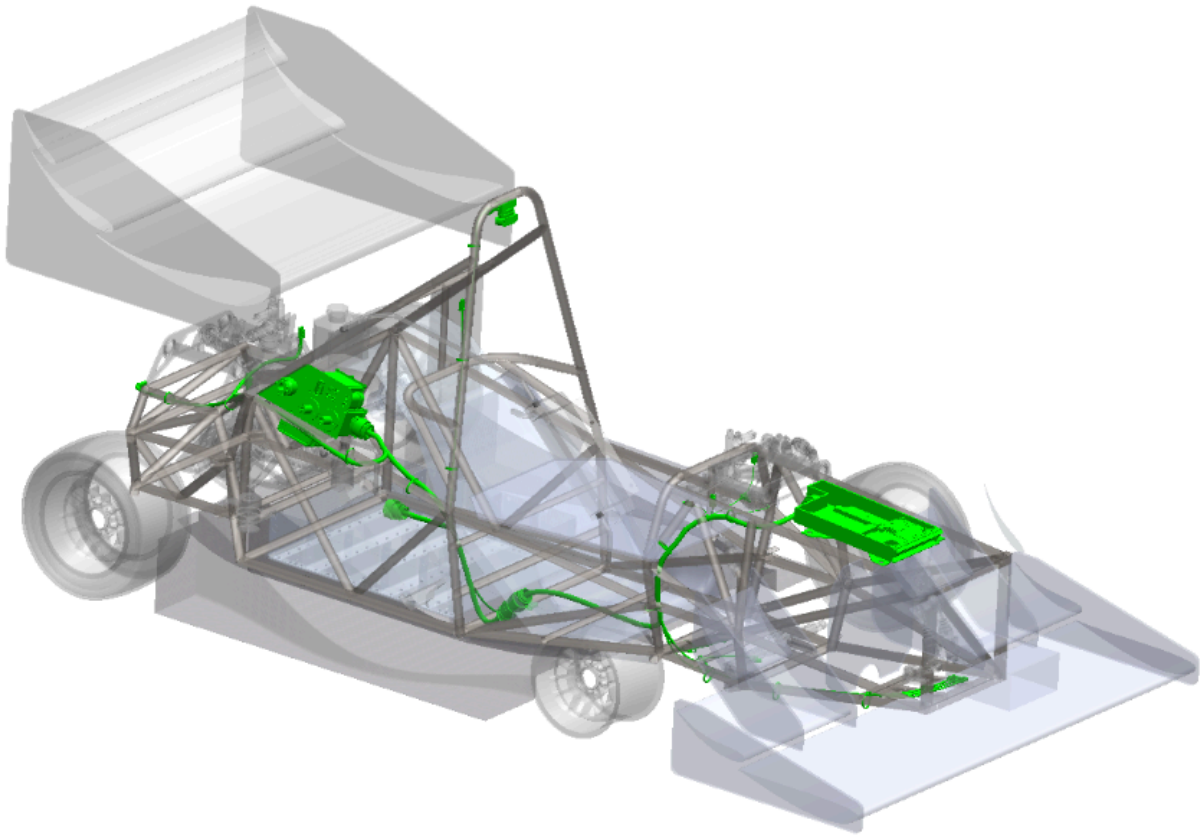


The BMS uses the NXP MC33771C as the main cell controller IC. I had chosen an STM32G474RE to monitor thermistor temperatures. The total tractive battery system is 2p140s, thus 10 modules were required and daisy chained through capacitive isolation TPL. Certain layout choices were made (clearances, isolation gaps, room placements) to make the board reliable.

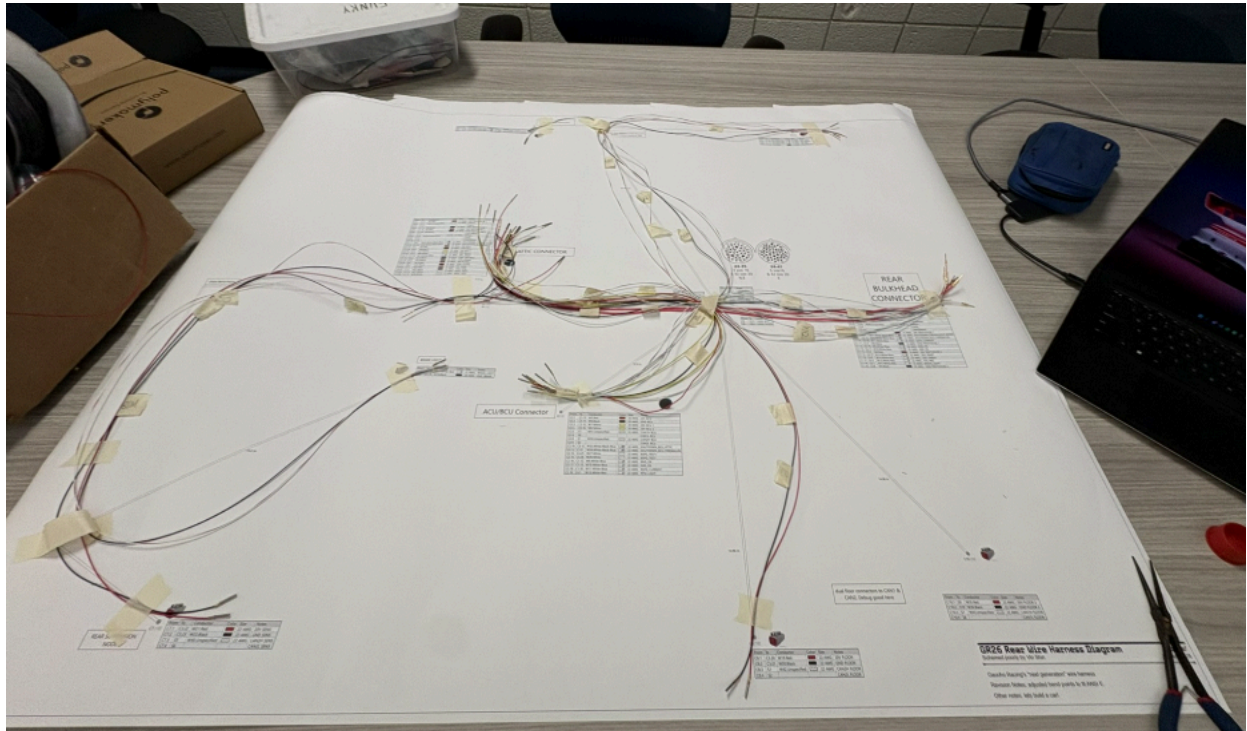
GR26 LV Electrical System

As an electronics lead for my university's Formula SAE team, Gaucho Racing, my role was the total design of the low voltage electrical system, which consisted of component integration, wire harnessing, low voltage power distribution, and certain control electronics.

[The design slides](#) (made in November 2025) give a general but outdated overview of the total electrical system and also the rationale behind certain changes.



The LV Electrical system is highlighted in green in contrast to GR26's assembly.



Wire Harness Manufacturing

CAN Digital Sensor Network - 2024

In my first year at UCSB, a group of 3 students and I received a \$2000 grant from UCSB Undergraduate Research and Creative Activities (URCA) to design and manufacture a custom sensor network for our Formula SAE Electric team's competition car, GR25. **Development and Testing of Sensor Deployment for Student Built Formula SAE Race Car**



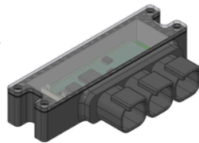
By Vin Shin, Shravya Salem Sathish, Anthony Ma, Tomi Chen
Advisor: Prof. Yogananda Isukapalli
Gaucha Racing, University of California, Santa Barbara, California 93106, USA



Overview

FSAE is an international collegiate design competition where students design, manufacture, and construct a race car (GR25) from scratch. Our project involved the design, manufacturing, and testing of a custom sensor network:

- **Dedicated Telecommunications Module**
- **Custom Ingress Protected Sensor Modules**
- **Manufacturing and Testing**



Data Processing and Collection

GR25 contains the Telecommunications Module (TCM) and the Engine Control Unit (ECU) that will collect or process the sensor suite's data.

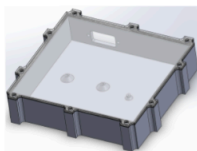
Torque Vectoring through the ECU

- Our vehicle implements a multi-motor configuration—Torque Vectoring can greatly increase performance.
- Algorithm requires data from wheel load, rear wheel speed, slip angle (DGPS), steering angle, and IMUs.



Data Collection through the TCM

- "Black Box" solution—Collects data of all the sensors and stores locally and through a SingleStore database.
- Consists of an Nvidia Orin Nano, CAN interface module, antennas, LTE router, and internal battery.

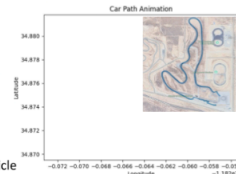


Data Collection and Analysis

We have tested various sensors and analyzed its data:

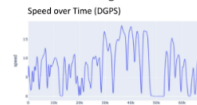
Differential GPS

- Tested on a real world track (Willow Springs International Raceway).
- Tested on a roadster car to simulate the expected environment of the GR25.



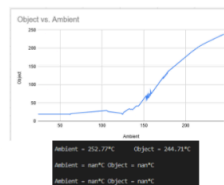
Slip Angle

- Calculated post-collection
- Determines the slip angle of vehicle based on heading



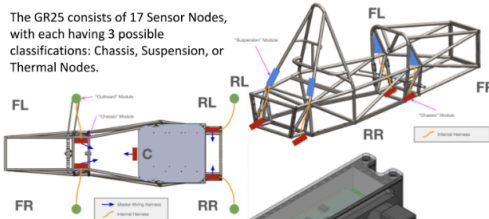
Thermal Nodes Stress Test

- Thermal Nodes will be placed within the upright which is capable of reaching temperatures of up to 300 F.
- Tested thermal capabilities through direct heat conduction to observe its performance under realistic conditions.
- Reached 250C (482 F) before failure.



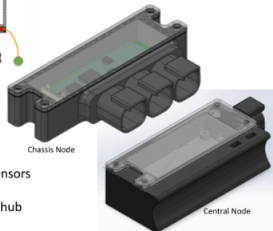
The Sensor Nodes

The GR25 consists of 17 Sensor Nodes, with each having 3 possible classifications: Chassis, Suspension, or Thermal Nodes.



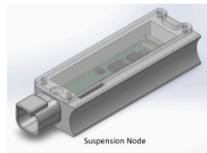
Chassis Nodes

- Located in 5 locations:
 - 4 nodes: Each corner near the wheels
 - Center of Chassis
- Contains IMU, Ride-Height sensors
- Differential GPS in center
- Acts as internal wire harness hub to other nodes.



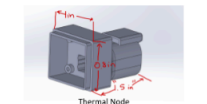
Suspension Nodes

- 4 Nodes placed close to suspension area.
- Contains potentiometer to measure suspension travel, Load Cell + conditioner for tyre load, and air speed sensor for aero data.



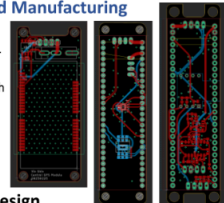
Thermal Nodes

- 8 nodes:
 - 4 with single-pixel probe for brake temp
 - 4 for tyre gradient
- Connected externally through secondary bus for safety.



Node Design and Manufacturing

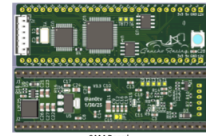
We custom-built nearly all of the nodes: the Sensor Array Module (SAM), Daughter PCBs, and the enclosures. This allows us full control and optimization of nodes such that it will be space-efficient and high performance when deployed within the vehicle.



PCB Design

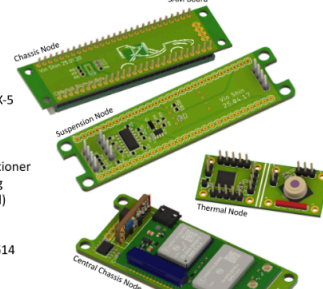
Sensor Array Module (SAM)

- STM32G474RE chip chosen for both intra-team compatibility and peripherals.
- Integrated STLINK for easy debugging and portability.
- Dual CANFD transceivers for CAN-Bus.
- 12V, 5V, and 3.3V supplies integrated on board for compatibility and ease.



Daughter PCBs

- Chassis:
 - BMI323 IMU
 - VL53LXX ToF
- Central Chassis:
 - InertialSense IG-2-IMX-5
 - Can Transceiver
 - Voltage Converter
- Suspension:
 - Load Cell signal conditioner
 - Potentiometer porting
 - MS4525DO (Air Speed)
- Thermal nodes:
 - Stacked Design
 - STM32 + MLX90640/614



Enclosure and Integration

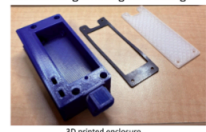
General Construction

- 3D printed two-part enclosure with standardized DTM 4-pin.
- Cut rubber gasketing for sealing.



Integration

- Molded around Chassis Tubes.
- Attached rigidly through singular bolt and zip tie.
- Takes minimal space within chassis.



Acknowledgements

Our team thanks URCA and our advisor Dr. Yogananda Isukapalli. We would also like to thank fellow experienced team members and leads of Gaucha Racing for insight and knowledge.